

**Department of Computer Applications  
UIT-RGPV (Autonomous) Bhopal**

**Subject code: MCA 108  
Semester: I**

**Subject: Operating system**

**For credits & marks refer your scheme**

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### **COURSE OUTCOME**

After completion of this course students will be able to:

1. Understand operating systems and evolution of OS.
2. Demonstrate an understanding of process management concepts and CPU scheduling algorithms.
3. Establish a comprehensive understanding of memory management techniques.
4. Know Inter process communication, synchronization techniques and Deadlock.
5. Understand File system, Storage system and Disk scheduling.

### **COURSE CONTENTS**

#### **UNIT I**

**Introduction:** Evolution of operating systems, Types of operating systems, Multitasking, Timesharing, Multithreading, Multiprogramming and, Real time operating systems, Different views of the operating system, System Programmer's view, User's view, Operating system concepts and structure, Layered Operating Systems, Monolithic Systems.

#### **UNIT II**

**Processes:** The Process concept, The process control block, Systems programmer's view of processes, Operating system services for process management, Scheduling algorithms, First come first serve, Round Robin, Shortest run time next, Highest response ratio next, Multilevel Feedback Queues, Performance evaluation of scheduling algorithms stated above

#### **UNIT III**

**Memory Management :** Memory management without swapping or paging, Concepts of swapping and paging, Page replacement algorithms namely, Least recently used, Optimal page replacement, Most recently used, Clock page replacement, First in First out (This includes discussion of Belady's anomaly and the category of Stack algorithms), Modeling paging algorithms, Design issues for paging system, Segmentation, Segmented Paging, Paged Segmentation

#### **UNIT IV**

**Inter-process Communication and Synchronization:** The need for inter-process synchronization, Concept of mutual exclusion, binary and counting semaphores, hardware support for mutual exclusion, queuing implementation of semaphores, Classical problems in concurrent programming, Dining Philosopher's problem, Bounded Buffer Problem, Sleeping Barber Problem, Readers and Writers problem, Critical section, critical region and conditional critical region, Monitors and messages.

**Deadlocks:** Concepts of deadlock detection, deadlock prevention, deadlock avoidance. Banker's Algorithm

#### **UNIT V**

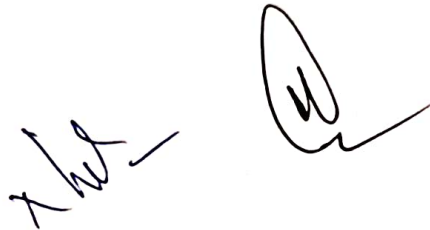
**File System:** File systems, directories, file system implementation, security protection mechanisms. Input/output: Principles of I/O Hardware: I/O devices, device controllers, direct memory access. Principles of I/O software: Goals interrupt handlers, device drivers, and device independent I/O software. User space I/O Software. Disks: Disk hardware, Disk scheduling algorithms (namely First come first



serve, shortest seek time first, SCAN, C-SCAN, LOOK and C-LOOK algorithms) Error handling, track-at-a-time caching, RAM Disks. Clocks: Clock hardware, memory-mapped terminals, I/O software.

**REFERENCE BOOKS:**

1. Milenkovic, M., "Operating Systems - concepts and Design" McGraw Hill International Edition Computer Science series 1992.
2. Galvin P., J.L. Abraham Silberschatz. "Operating System Concepts". John Wiley & Sons Company, 1989.
4. Tanenbaum, A.S. "Modern Operating System", Prentice Hall of India Pvt. Ltd. 1995.
3. William Stallings "Operating Systems", Prentice Hall of India Pvt. Ltd.

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**Department of Computer Applications  
UIT-RGPV (Autonomous) Bhopal**

**Subject code: MCA 109  
Semester: I**

**Subject: Programming and Problem solving in C**

**For credits & marks refer your scheme**

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**COURSE OUTCOMES:**

Upon successful completion of the course students will be able to:

1. Know basic terminologies of C programming.
2. Use of control statements, loops, I/O functions and array in C programming.
3. Know the use of Functions along with the call by value and call by reference
4. Understand dynamic memory allocation by the use of pointers and basic structure and Union.
5. Learn file handling functions and various preprocessor directives.

**Course content**

**Unit-1**

**Fundamentals of C Programming:** History of C; Structure of a C Program; Data types; Type modifiers; Constant & Variable, naming variables; Operators & expressions; Ternary operator; Type conversion & type casting; Priority & associativity of operators.

**Unit-II**

**Control Statements, Loops, I/O functions and Array:** Control Constructs – if-else, for, while, do-while; Case switch statement; Special constructs – Break, continue, exit(), goto & labels; Formatted & unformatted I/O-I/O Statements, Formatted I/O, String I/O Functions; Array-one dimension and multi dimension array.

**Unit-III**

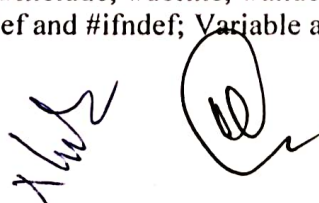
**Functions;** Arguments; Return value; Parameter passing – call by value, call by reference; Return statement; Scope, visibility and life-time rules for various types of variable, static variable; Calling a function; Recursion – basics, comparison with iteration, types of recursion-direct, indirect, tree and tail recursion, when to avoid recursion, examples; storage classes.

**Unit-IV**

**Pointers, Structure and Union** - & and \* operators, pointer expression, pointer arithmetic, dynamic memory management functions like malloc(), calloc(), free(); String; Pointer v/s array; Pointer to pointer; Array of pointer & its limitation; Function returning pointers; Pointer to function, Function as parameter; **Structure** – basic, declaration, membership operator, pointer to structure, referential operator, self referential structures, structure within structure, array in structure, array of structures; **Union** – basic, declaration; Enumerated data type; Typedef; command line arguments.

**Unit-V**

**Miscellaneous Features:** File handling and related functions; printf & scanf family; C preprocessor – basics, #Include, #define, #undef, conditional compilation directive like #if, #else, #elif, #endif, #ifdef and #ifndef; Variable argument list functions.

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**ReferenceBooks:**

1. Kerninghan & Ritchie "The C programming language", PHI
2. Schildt " C: The Complete reference " 4<sup>th</sup> ed. TMH.
3. Kanetkar Y. "Let us C", BPB.
4. Kanetkar Y.: " Pointers in C", BPB
5. Gottfried:" Problem Solving in C", Schaum Series
6. Jones, Harrow Brooklish " C Programming with Problem Solving ",Wiley Dream tech India.



**MCA I Semester**  
**Paper: MCA110**  
**Mathematical Foundation**

**Unit-I**

**Set Theory, Relation, Function:**

Set Theory: Definition of sets, countable and uncountable sets,

Relation: Definition, types of relation, composition of relations, Equivalence relation, Partial ordering relation.

Function: Definition, type of functions, inverse function, composition of functions, recursively defined functions, pigeonhole principle. Mathematical induction.

**Unit-II**

**Algebraic Structures:**

Definition, Properties, types: Semi Groups, Monoid, Groups, Abelian group, properties of groups, Subgroup, cyclic groups, Cosets, factor group, Permutation groups, Normal subgroup, Homomorphism and isomorphism of Groups, example and standard results, Introduction to Rings and Fields.

**Unit-III**

**Propositional Logic:**

Proposition, First order logic, Basic logical operation, truth tables, tautologies, Contradictions, Algebra of Proposition, logical implications, logical equivalence, predicates, Universal and existential quantifiers.

**Unit-IV**

**Graph Theory:**

Introduction and basic terminology of graphs, Planer graphs, Multigraphs and weighted graphs, Isomorphic graphs, Paths, Cycles and connectivity, Shortest path in weighted graph, Introduction to Eulerian paths and circuits, Hamiltonian paths and circuits, Graph coloring, chromatic number, Isomorphism and Homomorphism of graphs.

**Unit V**

**Matrices and Eigenvalue Problems:**

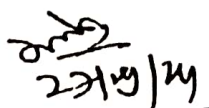
Rank of Matrix, Echelon form & Normal Form, Consistency of system of Linear Equations and their solutions, Eigen Values and Eigen Vectors, Cayley-Hamilton Theorem and its application to find the Inverse.

**Reference Books:**

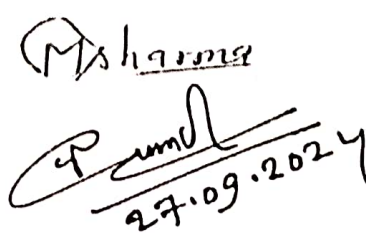
1. B.S.Grewal, *Higher Engineering Mathematics*, Khanna Publishers.
2. J.P.Trembley & R.P.Manohar, *Discrete Mathematical Structure with Applications to Computer Science*,
3. Kenneth H. Rosen, *Discrete Mathematics & its Applications*, Mc Graw Hill Publication.
4. Narsingh Deo, *Graph Theory With Applications To Engineering And Computer Science*, PHI
5. Kolman, Busby & Ross, *Discrete Mathematical Structures*, Pearson
6. A R Vasishtha and A K Vasishtha, *Modern Algebra*, Krishna Prakashan.



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Department of Computer Applications  
UIT-RGPV (Autonomous) Bhopal

Subject code: MCA 111  
Semester: I

Subject: Fundamentals of Computer Technologies

For credits & marks refer your scheme

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### COURSE OUTCOME

After completion of this course students will be able to:

1. Understand Digital computer architecture, memory and storage technologies.
2. Apply Number systems, and logic gates, along with techniques for Boolean function.
3. Overview of System software and Software tools,
4. Learn Data communication concepts.
5. Understand Cryptography & Information Security concepts

### COURSE CONTENTS

#### UNIT-I: Digital Computer & Storage devices

Model of a digital computer, Historical evolution of computers, Classification & Types of computers, Human being v/s Computer, Input output devices, Computer Application in E-business, Basic concepts of IT, concepts of Data & Information, Basic Structure of computer-CPU, Register & its types, ALU, Bus Architecture, Memory Program Counter, Register transfer language, Addressing Modes

**MEMORY AND STORAGE:** Memory Basics, the RAM, the ROM, the Flash Memory, Magnetic and Optical Storage

#### UNIT-II: Digital Systems

**Number system and codes:** Decimal Numbers, Binary Numbers, Decimal to Binary Conversions, Binary Arithmetic, 1's and 2's complements of Binary Numbers, Signed Numbers, Arithmetic Operations with Signed numbers, Hexadecimal Numbers, Octal Numbers, Digital Codes, Error Detection Codes.

**Logic Gates:** The Inverter, The AND gate, The OR gate, The NAND gate, NOR gate, The Exclusive-OR gate and Exclusive-NOR gate, Simplification of Boolean functions, Karnaugh map methods, SOP minimization

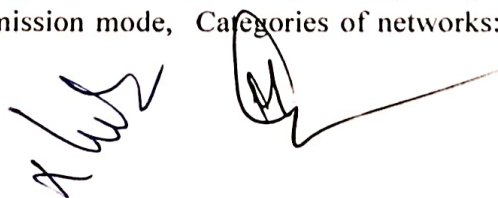
#### UNIT-III: Software systems

Introduction to software, Software Hierarchy, software Tools, Translator, Compiler: Phases of Compiler, Grouping of phases, Types of Compiler, Bootstrapping & Porting, Assembler: Elements of assembly language programming.

Assembler design Processor: Single pass assembler & Two pass assembler, Macro processor, Interpreters & its Benefits, Linker & Loaders.

#### UNIT-IV: Data Communication

**Data Communication:** Components, data representation Serial & Parallel transmission, Modes of data transmission, Multiplexing: FDM, TDM, WDM, Line Encoding: Unipolar, Polar, Bipolar Line configurations, Topology, Transmission mode, Categories of networks: LAN, MAN and



WAN, Inter networks, Intranet and Internet, Functions of the layers. Transmission media: Guided media – Unguided Media, Switching: Circuit switching, packet switching (virtual circuit and datagram approach), message switching

#### **UNIT-V: Network security**

**Introduction to Cryptography:** Security Terminologies and Principle, Security Threats, Types of attacks, Trojans and Backdoors, Viruses and Worms & its types, Phishing, Cryptography, Classical Cryptosystem, Cryptanalysis on Substitution Cipher (Frequency Analysis), Block Cipher, Data Encryption Standard (DES), Triple DES, Digital Signature,

**Information Security:** Firewalls – Design and Types of Firewalls, Email Security: Services Security for Email Attacks Through Emails, Pretty Good Privacy (PGP), S-MIME.

#### **REFERENCE BOOKS:**

1. Morris Mano: Computer System Architecture, PHI.
2. Malvino & Leach, "Digital Principles and Applications", TMH.
3. M. Morris Mano, "Digital Logic Design", PHI
4. "System Software": D M Dhamdhare, TMH
5. Data Communication & Networking – B.A. Forouzan, Tata Mc-Graw Hill
6. Cryptography and Network Security – by Atul Kahate – TMH.

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## UIT-RGPV (Autonomous) Bhopal

Subject code- MCA 112

Subject: Technical Communication

Semester: I

For credits & marks refer to scheme

At the end of the course learners are expected to:

| S. No | Course Outcome                                           |
|-------|----------------------------------------------------------|
| CO-1  | Develop effective communication skills for professionals |
| CO-2  | Analyze, interpret the technical content effectively     |
| CO-3  | Improve Writing skills.                                  |
| CO-4  | Enhance skills of Business Correspondence                |
| CO-5  | Foster public speaking and presentation skills           |

### COURSE OBJECTIVE

Exploring concepts of communication and developing analytical reading skills. Analyzing and interpreting textual content effectively and developing writing skills.

#### UNIT 1

Communication: Basics of Technical communication, Process of communication, Levels of communication, Flow of Communication, Types of Communication- verbal, Non-verbal, written, Barriers to Communication, Essentials of Effective communication, Cross-cultural variations.

#### UNIT 2

Analytical Reading Skills: Improving Comprehension Skills, Reading Difficulties, Active and Passive Reading, Benefits of Effective Reading, Reading Comprehension, Multiple Choice questions based on unseen passages. Analyze, interpret the textual content effectively

#### UNIT 3

Developing Writing Skills: Elements of Effective writing, Writing Effective sentences, Developing logical paragraphs, The Art of Condensation, Steps to Effective Précis Writing, Essay writing, Writing for the Web, Leveraging AI tools.

#### UNIT 4

Business Correspondence - Business Letters, Parts and Layouts of Business Letters, Writing Resume, Letter of Application, Calling/ Sending Quotations/ Orders/ Complaints. Writing Emails, advantages and limitations of E- Mails, Email Etiquette.

## **UNIT 5**

Speaking and listening skills. Effective speaking and listening skills, The art of conversation and discussion, Debate, Preparing and Delivering Effective Presentations, Using Visual Aids and Multimedia Tools for Effective Presentation.

### **Reference Texts**

1. Rizvi, M Asraf. **Effective Technical Communication** New Delhi: Tata McGraw-Hill, 2005
2. Raman, Meenakshi and Sharma, Sangeeta . **Technical Communication- Principles and Practice**. New Delhi: Oxford University Press, 2004.
3. Hemamalini, Dr. M. **Technical Communication: Principles and Practice**. New Delhi: Oxford University Press, 2015.
4. Kumar, Sanjay and Pushp Lata. **English for Effective Communication**. New Delhi: Oxford University Press, 2015
5. Lakshminarayan, K .R. and Murugavel, T. **Communication Skills for Engineers**, India: Scitech Publications (India) Pvt Limited.2008.